Ofqual QAN	**600/0406/X**
Quartz ID	**1156, 1157**
Qualification	**Level 4 Diploma in Computing. Level 4 Diploma in Computing with Business Management**

Assessment	**Algorithms and Mathematical Concepts for Computing**
Assessment Type	**Examination**
Date	**Wednesday 28th May 2025**
Time	**2.00pm – 5.00pm**

Examination Paper

Answer ALL questions.

Clearly cross out surplus answers.

Time: 3 hours

The maximum mark for this paper is 100.

Any reference material brought into the examination room must be handed to the invigilator before the start of the examination

Answer ALL questions

Marks

Question 1

This question relates to the Software Development Process and Algorithm:

a) During the early stages of the software development lifecycle, a phase focuses on understanding the problem domain, identifying stakeholder needs and defining what the system should achieve. What is the name of this phase? **1**

b) Which stage of the software development lifecycle focuses on addressing bugs, implementing improvements, and providing ongoing support for the system? **1**

c) i) Define a flowchart **and** explain its purpose in software development. **2**

ii) Compare flowcharts with pseudocode. **2**

iii) Write pseudocode to display a greeting message five times using a loop. Use selection so that the message for the first time (i.e. when the counter is 1) is different from the rest (e.g., display “Welcome!” the first time and “Hello again!” for the remaining times). After the greetings have been displayed five times, display a final message such as “Session over.” **4**

Total 10 Marks

© NCC Education Limited 2025

Question 2

This question relates to numbers and number systems:

a) Which of the following are true about number systems? **2**

a) In a binary system, the digits are limited to the values 0 and 1.
b) The decimal system uses base 8.
c) The hexadecimal system is often used in computer memory addressing because it represents values in base 10.
d) In the octal system, each digit can range from 0 to 7.

b) Define the greatest common divisor (GCD) **and** explain its significance in number theory. **2**

© NCC Education Limited 2025

c) Perform the following conversions **and** show your working. **2**

i) Convert the decimal number 91 to binary. Show your working.

ii) Convert the binary number 1101 to octal. Show your working. **2**

iii) Convert the hexadecimal number 2F to decimal. Show your working. **2**

Total 10 Marks

Question 3

Answer the following questions related to logic and Boolean algebra:

a) Which of the following symbols represents the logical NOT operation? **1**

i) $\wedge$ **ii)** $\neg$

iii) $\rightarrow$ **iv)** $\vee$

b) Complete the truth table for the Boolean expression: **4**

$$(P \wedge Q) \vee (\neg P \wedge Q)$$

P	**Q**	**P ∧ Q**	**¬P**	**(¬P ∧ Q)**	**(P ∧ Q) ∨ (¬P ∧ Q)**
F	F				
F	T				
T	F				
T	T				

[Insert the marking scheme for part (b) here.] ***[Insert***

c) Write and explain the concept of the First De Morgan's Theorem **and** provide ONE (1) example. Include the truth table for the theorem. **5**

[Insert the marking scheme for part (b) here.] ***[Insert***

Total 10 Marks

© NCC Education Limited 2025

Question 4

Answer the following questions related to Set Theory:

a) Which of the following represents the Cartesian Product of two sets? **1**

i) $A \cap B$ **ii)** $A \times B$

iii) $A \cup B$ **iv)** $A \subset B$

[Insert the marking scheme for part (b) here.]

b) Define the concept of a subset **and** provide an example. **2**

[Insert the marking scheme for part (b) here.]

c) Find the complement of the set B = {2, 4, 6} relative to the universal set U = {1,2,3,4,5,6,7} **and** explain the process. **2**

[Insert the marking scheme for part (b) here.]

d) Define the term "intersection of sets" and find the intersection of A = {1, 2, 3}, B = {2, 3, 4}, and C = {3, 4, 5}. Show the intersections of two sets at a time. **3**

[Insert the marking scheme for part (b) here.]

e) Explain the difference between a finite set and an infinite set with an example for each. **2**

[Insert the marking scheme for part (b) here.]

[Insert the marking scheme for part (b) here.]

Total 10 Marks

© NCC Education Limited 2025

Question 5

Answer the following questions related to statistics:

a) Explain the concept of standard deviation. Provide **ONE (1)** example where it is useful in data analysis. **2**

b) Write the formula for standard deviation **and** calculate the standard deviation for the dataset: 3, 7, 5, 9, 6. Show your working. **3**

c) Define the term "skewness" in statistics. Explain the difference between positive **and** negative skewness. **2**

d) Define mode. **3**
Find the mode of the dataset: 2, 3, 3, 5, 7, 7, 7, 8, 9.
Provide an example of when the mode is useful.

Total 10 Marks

© NCC Education Limited 2025

Question 6

Answer the following questions related to Data Structures and Arrays:

a) Which of the following operations is **NOT** typically supported by a stack? **1**

i) Push
ii) Pop
iii) Peek
iv) Dequeue

b) Define a "linked list," describe **TWO (2)** primary operations performed on a linked list **and** provide an example application of a linked list. **4**

c) Write a function in pseudocode to insert a node at the beginning of a singly linked list. **5**
You can assume the existence of the function createNode(x) which creates a new node containing data (x), returning a pointer to the node.

Total 10 Marks

Question 7

Answer the following questions related to Data Structures:

a) What is the maximum number of elements that can be stored in an array of size *n* at a given index? **1**

i) n
ii) n/2
iii) 1
iv) $2^{(n+1)}$

b) Define the term "linked list" and explain its key property. **2**

c) Explain the difference between a stack and a queue in terms of data access order. Mention typical use cases for each. **3**

© NCC Education Limited 2025

d) Write pseudocode to push an element onto a stack. Complete the function body below. Make sure the case where the stack is full is handled. **4**

```
Function push(stack, value):
    [BODY]
End Function
```

Total 10 Marks

Question 8

Answer the following questions related to algorithm efficiency and searching techniques:

a) Which of the following sorting algorithms has the worst-case time complexity of $O(n^2)$? **1**

i) Quick Sort

ii) Merge Sort

iii) Bubble Sort

iv) Heap Sort

b) Describe the concept of 'divide and conquer' as used in sorting algorithms, with an example of a sorting algorithm that uses 'divide and conquer'. **2**

c) Write pseudocode for the Bubble Sort algorithm. Complete the body of the function below: **4**

```
Function bubbleSort(arr):
    [BODY]
End Function
```

d) Explain the difference between Merge Sort and Quick Sort in terms of their time complexities. Which data structure is more suitable for each algorithm.? **3**

Total 10 Marks

© NCC Education Limited 2025

Question 9

Answer the following questions related to sorting algorithms and recursion:

a) Which of the following sorting algorithms repeatedly compares adjacent elements and swaps them if they are in the wrong order, progressively moving elements toward their correct positions? **1**

i) Bubble Sort
ii) Quick Sort
iii) Selection Sort
iv) Merge Sort

b) Explain selection sort and bubble sort in terms of how they compare and swap elements. After explaining, outline the key difference between the operations of selection sort and bubble sort regarding comparing and swapping elements in one sentence. **3**

c) Fill in the blanks in the pseudocode for computing the nth Fibonacci number using recursion: **4**

```
Function fibonacci(n):
  if n <= _____:
     return _____
  return _____ + _____ (n - 2)
End function
```

d) Explain how recursion can be used to calculate the factorial of a number. **2**

Total 10 Marks

Question 10

Answer the following questions related to tree data structures:

a) Which of the following is **TRUE** about a binary tree? **1**

i) A binary tree always has exactly two children per node.
ii) The root of the binary tree must have the smallest value.
iii) A binary tree can have any number of children per node.
iv) A binary tree can have at most two children per node, but not necessarily both.

b) In a binary search tree (BST), for any given node, all the values in its _____ subtree are smaller, and all the values in its _____ subtree are greater. **1**

© NCC Education Limited 2025

c) Construct a binary tree for the following sequence of numbers: 10, 5, 15, 2, 8, 12, 20. Ensure that 10 is the root. **4**

d) Using the binary tree constructed in Question 10c, explain the steps to search for the value 8. Describe each comparison and movement until the value is found or determined to be absent. **4**

Total 10 Marks

End of paper

© NCC Education Limited 2025